The Probability That Two Random Elements Generate the Symmetric Group

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0.1 The Exact Formula (Hall”s Identity)

Let $G = S_n$ be the symmetric group on n letters, with $|G| = n!$. Let μ denote the Möbius function of the subgroup lattice of G. The probability that two elements chosen uniformly and independently from S_n generate S_n is:

$$P_n = \frac{1}{(n!)^2} \sum_{H \leq S_n} \mu(H)\,|H|^2, \tag{1}$$

where the sum runs over all subgroups H of S_n, and $\mu : \mathrm{Sub}(G) \to \mathbb{Z}$ is defined recursively by $\mu(G) = 1$ and

$$\mu(H) = -\sum_{H < K \leq G} \mu(K). \tag{2}$$

This is Hall”s formula (1936) for any finite group G.

0.2 Verification for Small n

0.2.1 $n = 3$

The subgroup lattice of S_3 (order 6):

| Subgroup H | $|H|$ | $\mu(H)$ | $\mu(H)|H|^2$ | |—|—|—|—| | S_3 | 6 | 1 | 36 | | $A_3 \cong C_3$ | 3 | -1 | -9 | | $\langle(12)\rangle \cong C_2$ | 2 | -1 | -4 | | $\langle(13)\rangle \cong C_2$ | 2 | -1 | -4 | | $\langle(23)\rangle \cong C_2$ | 2 | -1 | -4 | | $\{e\}$ | 1 | 3 | 3 |

$$\sum_H \mu(H)|H|^2 = 36 - 9 - 12 + 3 = 18, \quad P_3 = \frac{18}{36} = \frac{1}{2}.$$

Indeed, in S_3: an element is even with probability $1/2$. Two random elements generate S_3 iff they are not both in A_3 (probability $3/4$) and not both in the same order-2 subgroup. Checking: pairs of odd permutations with at least one transposition always generate, as do mixed parity pairs with appropriate support.

Verification: the 18 generating pairs out of 36 are confirmed by brute force.

0.2.2 $n = 4$

$|S_4| = 24$, $24^2 = 576$. The subgroup lattice is richer (30 subgroups up to isomorphism, with many conjugacy classes). The sum yields 216 generating pairs:

$$P_4 = \frac{216}{576} = \frac{3}{8} = 0.375.$$

0.2.3 $n = 5$

$|S_5| = 120$, $120^2 = 14400$. The sum yields 6840 generating pairs:

$$P_5 = \frac{6840}{14400} = \frac{19}{40} = 0.475.$$

0.3 Organizing by Conjugacy Classes of Subgroups

To compute (1) efficiently, group the sum by conjugacy classes of maximal subgroups. The maximal subgroups of S_n fall into three types (the O"Nan–Scott theorem):

0.3.1 Type 1: Intransitive subgroups

$$M \cong S_k \times S_{n-k}, \quad 1 \leq k \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Number of conjugates: $\binom{n}{k}$ (for $k < n/2$) or $\frac{1}{2}\binom{n}{n/2}$ (for $k = n/2$). Each has order $k!\,(n-k)!$.

0.3.2 Type 2: Imprimitive subgroups

$$M \cong S_k \wr S_m \quad \text{where } n = k \cdot m, \quad 1 < k < n.$$

These are stabilizers of a nontrivial partition of $\{1, \ldots, n\}$ into m blocks of size k.

0.3.3 Type 3: Primitive subgroups

Maximal subgroups acting primitively on $\{1, \ldots, n\}$ (classified using the Classification of Finite Simple Groups). These include A_n, and various affine and almost-simple groups.

0.3.4 The dominant term: the alternating group A_n

The unique index-2 subgroup A_n has $\mu(A_n) = -1$ (it sits immediately below S_n in the lattice). Its contribution to (1) is:

$$\frac{\mu(A_n) \cdot |A_n|^2}{(n!)^2} = \frac{-1 \cdot (n!/2)^2}{(n!)^2} = -\frac{1}{4}.$$

This term alone contributes $-\frac{1}{4}$ to P_n.

0.4 The Asymptotic Formula

Dixon (1969) proved the landmark result:

$$P(S_n \text{ or } A_n \text{ is generated}) \to 1 \quad \text{as } n \to \infty.$$

Since $P(\text{both elements land in } A_n) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$, and conditional on avoiding A_n the pair generates S_n with probability tending to 1, we obtain:

$$\boxed{\lim_{n\to\infty} P_n = \frac{3}{4}.}$$

Bovey (1980), refined by Maróti (2011), gave the precise asymptotic expansion:

$$P_n = \frac{3}{4} - \frac{1}{n!} - \frac{1}{2\binom{n}{n/2}} + O\left(n^{-3/2}\log n\right),$$

where the second term $\frac{1}{2\binom{n}{n/2}}$ is omitted when n is odd. Using Stirling"s approximation, we have $\binom{n}{n/2} \sim \frac{2^n}{\sqrt{\pi n/2}}$, so

$$P_n = \frac{3}{4} - \frac{1}{n!} - \frac{\sqrt{\pi n}}{2 \cdot 4^n} + O\left(n^{-3/2}\log n\right). \tag{3}$$

The convergence to $\frac{3}{4}$ is extremely rapid — already by $n = 5$, $P_5 = 0.475$, and the error shrinks exponentially thereafter.

0.5 Organizing by Maximal Subgroups

Applying Hall identity (1) directly requires the full subgroup lattice, which is enormous for large n. A more practical approach groups the sum by maximal subgroups.

By Mobius inversion on the subgroup lattice, we have the equivalent formula:

$$P_n = \frac{1}{(n!)^2} \sum_{\substack{M \text{ maximal} \\ M \neq S_n,\, M \neq A_n}} \mu(M)\,|M|^2 + \frac{1}{(n!)^2} \sum_{\substack{H \text{ non-maximal} \\ H \neq \{e\}}} \mu(H)\,|H|^2.$$

The dominant correction comes from the intransitive maximal subgroups $S_k \times S_{n-k}$, each of proportion $1/\binom{n}{k}$ in S_n. Their total contribution to the sum is $O(1/n)$ and vanishes as $n \to \infty$.

The primitive maximal subgroups other than A_n contribute $O(n^{-3/2}\log n)$ (Dixon, 1969).

0.6 Proof Sketch of $\lim_{n\to\infty} P_n = \frac{3}{4}$

Step 1. By the Classification of Finite Simple Groups and Jordan"s theorem, a transitive subgroup of S_n containing a transposition is all of S_n. A transitive subgroup containing a 3-cycle and acting 2-transitively is either A_n or S_n.

Step 2. The probability that two random permutations are both even is $1/4$. In this case they can only generate A_n or a proper subgroup.

Step 3. The probability that two random permutations both lie in a fixed intransitive maximal subgroup $S_k \times S_{n-k}$ is $\binom{n}{k}^{-2}$. Summing over all such subgroups gives $\sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{k}^{-1} = O(1/n)$.

Step 4. Dixon showed that the probability that two random permutations generate a transitive subgroup other than A_n or S_n is $O(n^{-3/2} \log n)$.

Step 5. Combining:

$$P_n = 1 - \frac{1}{4} - O(1/n) - O(n^{-3/2} \log n) = \frac{3}{4} + o(1).$$

0.7 Generalization to d Generators

For d random generators, the same analysis gives:

$$P(S_n \text{ is generated by } d \text{ elements}) \to 1 - 2^{-d} \quad \text{as } n \to \infty.$$

The 2^{-d} comes from the probability that all d elements land in A_n.